Vikas Shukla

(716) 903 2501 | vikasofficial927@gmail.com | [LinkedIn] / [GitHub] / [Portfolio] | Richmond, VA

SUMMARY

Data Scientist with 7 years of experience crafting production ML systems and AI-powered workflows, turning complex processes into scalable, real-world impact.

EDUCATION

M.S. in Data Science (GPA: 3.7 / 4.0) | University at Buffalo *Feb 2022*

B.Tech. in Electrical & Electronics Engineering (GPA: 8.6 / 10) | SRM University, India *May 2014*

TECHNICAL SKILLS

Languages & Tools: OpenAI, Transformers, YOLO, DeepSORT, Elevanlabs, LlamaIndex, LangChain, OCR, Text-to-Speech, Python, PySpark, SQL, Linux, Numpy, Pandas, Sklearn, TensorFlow, Keras, PyTorch, Gensim, NLTK, Tweepy, OpenCV, Tableau, Visual Studio (VS) Code, Copilot, Claude, ChatGPT, Gemini, Hugging Face, Agents, Prompt Engineering, Fine-Tuning (LoRA, PEFT), RLHF, DPO, Tool Calling, GPT, BERT, Causal Inference, React, Gradio, Langfuse, Streamlit

Databases & Warehouses: MYSQL, PostgreSQL, Hive, Snowflake, MongoDB, Elastic-Search, Solr, BigQuery, Redshift, DynamoDB, S3

Services: AWS, MS Azure, Google Cloud Platform (GCP), MapReduce, GitHub, Bitbucket, Jenkins, Docker, Kubernetes, JIRA, Yellowbrick

EXPERIENCE

Sr. Data Scientist | **Capital One** | *Richmond, VA* *Oct 2024 – Present*

- Built automated Python/SQL data quality monitoring system with real-time KPI alerting, catching data issues before business impact in ~80% of cases and reducing weekly metric variance by 20%.
- Consolidated 40M+ transactions across disparate systems into a unified self-serve analytics layer, eliminating 60% of manual analysis cycles.
- Productionized fraud scoring models with automated threshold management via CI/CD pipeline, achieving 98% model adoption across Risk Management teams.
- Partnered with engineering and compliance to align AI model outputs with U.S. regulatory standards; documented diagnostics and root-cause analysis fixes.
- Executed Risk Management and Audit testing using Test of Design and Test of Effectiveness for data pipeline controls. Mapped controls to risks and
- KRIs, reconciled populations, sampled and re-performed checks, and produced traceable audit evidence.
- Built an XGBoost risk forecasting model on governance KRIs across completeness, timeliness, and lineage to predict compliance trends and workload. Improved forecast accuracy by 16 percent. Added reason codes for explainability and set monitoring and thresholds to support reviews and audits.
- Introduced ***Debt-Sensitive Risk Score*** for millions of transactions using an in-house ***XGBFit*** technique to identify customers who are likely to incur higher debt $Losses in the future by ramping-up their $Balance on cards; followed by collaborating with Business Partners to explore its use-cases in Credit Extension, Auth Decline decisions.
- Drove an Incremental Annual Profit of **$20 Million** by assigning *Pay Over Time Revolving Limit* for 500K Small Business Charge Card Applicants via deploying Line-Sensitive Spend, Revolve & Write-off models developed using in-house ***UGBFit*** technique.
- Led the end-to-end deployment of Credit Line models during the CMV Globalization (CMV 2.0) initiative to generate **~$80 MM Annual Profit** by syncing with Tech, Governance, Product & Strategy partners.

Data Scientist | **Bayer** | *St. Louis, MO* *Jul 2023 – Oct 2024*

- Built XGBoost models for complex trial and breeding datasets with feature engineering, repeated cross-validation, and hyperparameter tuning, improved predictive accuracy and reduced model iteration time from 3 days to 1 day
- Containerized end-to-end analytics environment in Docker, rebuilt RMarkdown reporting pipelines, and eliminated 90% of reporting failures through automated job scheduling.
- Operationalized an R Shiny web application for interactive ML model training and real-time visualization, adding logistic regression with stratified cross-validation, ROC/PR curves, and live threshold tuning; improved AUC from

0.78 to 0.86 and F1 by 14%.
- Built predictive models for complex trial and breeding data using Gradient Boosting and regularized Logistic Regression with repeated cross-validation; cut model iteration time from 3 days to 1 day.
- Authored complex data analysis reports in LaTeX with statistical plots and model performance tables for scientific and business stakeholders.

Software ML Engineer | Amazon | *Bellevue, WA* *Jun 2022 – Apr 2023*
- Designed and deployed secure notebook infrastructure for ML model training and deployment, reducing model development-to-production time by 40% for the Alexa Data Science team.
- Built AI-powered notification quality assessment framework integrating user settings, reachability signals, and A/B test results across push, SMS, and email channels; improved Alexa user engagement and retention by 20%.
- Launched uplift-based targeting system with guardrails against notification fatigue, reduced total sends by ~22% with no engagement loss, cut SMS/email failures by ~30%, and lifted incremental conversion by ~8%.
- Implemented automated business logic for A/B, multivariate, and auto-targeting marketing experiments using Python, increased campaign ROI and strategy effectiveness.

Data Scientist | SGS Tekniks | *Gurugram, India* *Nov 2014 – May 2018*
- Built Linear Regression forecast models for customer order prediction, improving AUC by 10% and reducing RMSE by 12% for date prediction and 5% for quantity prediction.
- Automated Python/SQL data ingestion pipelines for orders, tickets, and marketing data with schema validation and daily model retraining; improved data freshness from weekly to daily, saving ~25 analyst hours/month.
- Built marketing experiment framework with power analysis, CUPED variance reduction, and uplift modeling; delivered 7% lift in incremental conversion at flat spend, narrowing confidence intervals by ~25%.

KEY PROJECTS

IntelliQuery Engine (LLM + RAG Agent) GitHub | *Python, FastAPI, FAISS, React Native, SQL*
- Built retrieval-augmented generation system using LLM embeddings and FAISS vector search enabling semantic queries across 10K+ document chunks with <800ms response time. Reduced unsupported responses by ~60%.
- Implemented intent-routing with lightweight classification for finance/legal queries, improving accuracy by ~45% and cutting manual validation effort by ~50%.
- Deployed scalable FastAPI backend integrated with React Native application (caseFit) delivering production-grade low-latency AI responses.

Fraud Detection System GitHub | *Python, XGBoost, SQL*
- Trained XGBoost classifier with target encoding and transaction pattern features; validated with decile analysis, Lorenz curve, and Gini coefficient. Top 2 deciles captured majority of fraud with high recall.

Cancer Gene Detection GitHub | *R, Python, SQL*
- Reduced 20K+ genomic features to 500 via PCA (99.7% reduction)
- Applied clustering algorithms on principal components and validated patient clusters against ground truth labels.

LLM-Generated Football Commentary System [GitHub/Download Demo Video] **|** *YOLOv8, DeepSORT, Kalman Filter, EasyOCR, OpenAI ChatGPT, ElevenLabs*
- Built an AI pipeline for real-time Football commentary using YOLOv8 for player & ball detection, DeepSORT for multi-player tracking with identity mapping and a Kalman Filter for stable ball tracking.
- Improved jersey number recognition accuracy using EasyOCR with preprocessing steps like CLAHE, unsharp masking & upscaling.
- Developed event detection for ball passes by analyzing ball trajectory, player proximity & motion history patterns.
- Generated human-like live audio commentary by combining OpenAI's GPT based text generation with ElevenLabs voice synthesis.